

CONVERSION REFERENCES

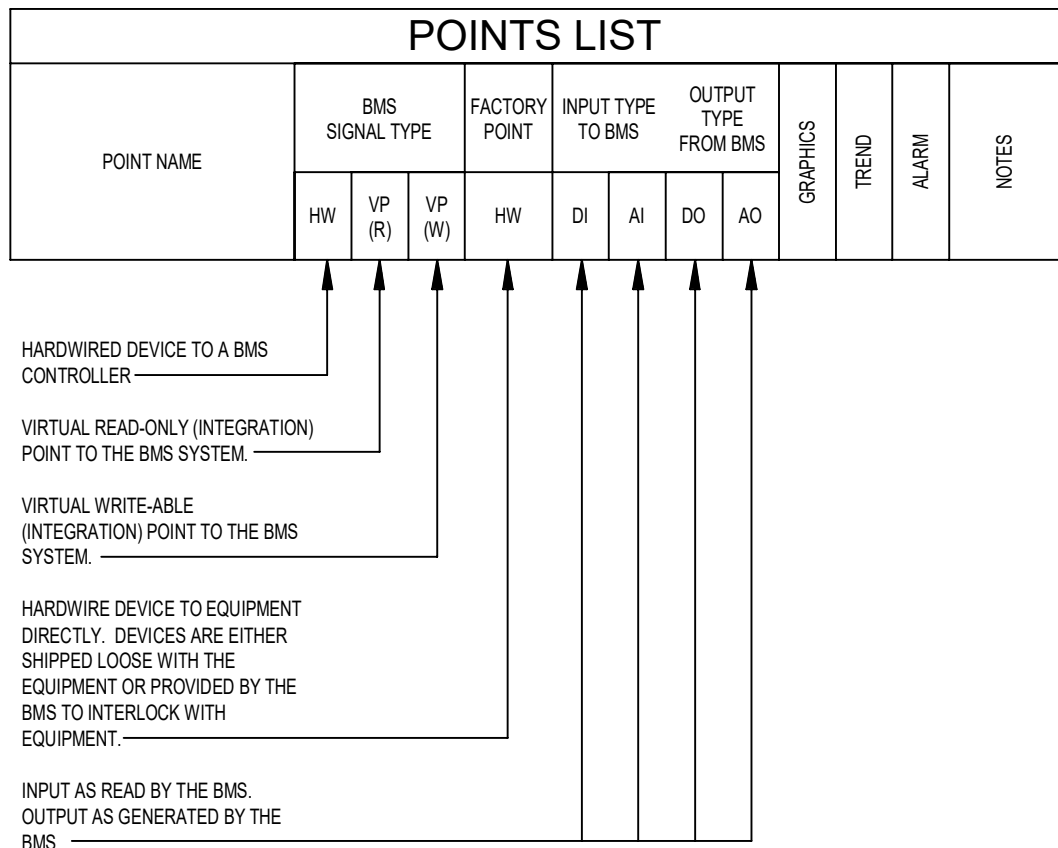
°C	°F	Meter	Feet
-20	-4	0.25	0.82
-18	-0.4	0.5	1.64
-16	3.2	0.15	2.46
-14	8.8	1	3.28
-12	10.4	1.25	4.1
-10	14	1.5	4.92
-8	17.6	1.75	5.74
-6	21.2	2	6.56
-4	24.8	2.25	7.38
-2	28.4	2.5	8.2
0	32	2.75	9.02
2	35.6	3	9.84
4	39.2	3.25	10.66
6	42.8	3.5	11.48
8	46.4	3.75	12.3
10	50	4	13.12
12	53.6	4.25	13.94
14	57.2	4.5	14.76
16	60.8	4.75	15.58
18	64.4	5	16.4
20	68	5.25	17.22
21	69.8	5.5	18.04
22	71.6	5.75	18.86
23	73.4	6	19.68
24	75.2	6.25	20.5
25	77	6.5	21.32
26	78.8	6.75	22.14
27	80.6	7	22.96
28	82.4	7.25	23.78
29	84.2	7.5	24.6
30	86		
31	87.8		
32	89.6		

°F = (°C) x 1.8 + 32
1 Meter = 3.28 Feet

GENERAL BMS NOTES

- THOROUGHLY REVIEW THE ENTIRE SET OF CONTRACT DOCUMENTS (E.G. PLANS, SPECIFICATIONS, ETC.) AS A WHOLE.
- CONTROL DRAWINGS DO NOT INCLUDE ALL HVAC CONTROL DEVICES. PACKAGED MECHANICAL EQUIPMENT ARE FURNISHED WITH FACTORY-MOUNTED AND WIRED CONTROL WHICH ARE TO BE INCORPORATED INTO THE BMS. DRAWINGS ARE DIAGRAMATIC SHOWING GENERAL INTENT FOR BMS. PROVIDE FUNCTIONAL CONTROL SYSTEM INTERFACE WITH LIDL NATIONWIDE BMS.
- MOTOR STARTERS MAY NOT BE ADJACENT TO ITS EQUIPMENT. REFER TO ELECTRICAL DRAWINGS FOR ACTUAL LOCATION.
- WORK CLOSELY WITH VFD START-UP PERSONNEL TO MEET FACILITY REQUIREMENTS. PRIOR TO SUBSTANTIAL COMPLETION.
- PROVIDE SUBMITTALS FOR THE BMS DRAWINGS INCLUDING FLOOR AND ROOF PLANS CALLING OUT CONTROL DEVICES, WIRING NETWORK, MONITORING DEVICES, AND CONTROL POINT MATRIX FOR REVIEW BY EOR, AOR, AND/OR OWNER REPRESENTATIVE AND FOR RECORD PURPOSES. ELECTRONIC AUTOCAD DWG FILES AND/OR REVIT FILES ARE AVAILABLE UPON REQUEST.
- PROVIDE A RECORD SET OF AS-BUILT DRAWINGS. ELECTRONIC AUTOCAD DWG FILES AND/OR REVIT FILES ARE AVAILABLE UPON REQUEST.
- CONSULTANT DRAWINGS INCLUDING BUT NOT LIMITED TO STRUCTURAL, MECHANICAL, ELECTRICAL, PLUMBING, REFRIGERATION, FIRE PROTECTION, AND CIVIL ARE SUPPLEMENTARY TO THE DRAWINGS. NOTIFY THE ARCHITECT OF ANY AND ALL DISCREPANCIES IDENTIFIED BETWEEN THE CONSULTANTS DRAWINGS WITH A WRITTEN REQUEST FOR CLARIFICATION. WORK INSTALLED IN CONFLICT WITH THE DRAWINGS AND PROJECT MANUAL SHALL BE CORRECTED BY THE CONTRACTOR AT THE CONTRACTORS EXPENSE.
- CONTROLS CONTRACTOR SHALL WORK CLOSELY WITH BALANCING CONTRACTOR TO MEET FACILITY REQUIREMENTS. PRIOR TO SUBSTANTIAL COMPLETION.
- ALL CONTROL, CONDUIT AND WIRE SHALL NOT BE INSTALLED SUCH THAT MECHANICAL MAINTENANCE ARE PROHIBITING OR LIMITED.
- CONTROLS CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING CONDUIT, WIRE, AND TRANSFORMERS TO PROPERLY ENERGIZE CONTROLLERS, ACTUATORS, AND SENSORS (I.E. HUMIDITY, CO2, PRESSURE, ETC).
- WHERE A DISCREPANCY AMONG SPECIFICATION, SEQUENCE, AND SCHEMATICS OCCURS, THE MOST STRINGENT SHALL APPLY.

POINTS LIST KEY



POINTS LIST KEY: EXAMPLES ONLY

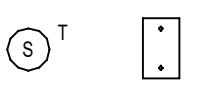
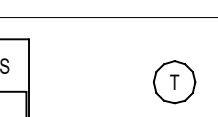
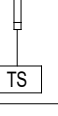
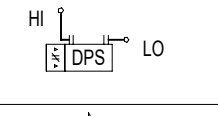
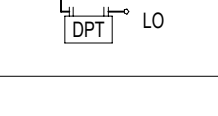
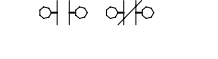
POINTS LIST										
POINT NAME	BMS SIGNAL TYPE		FACTORY POINT	INPUT TYPE TO BMS		OUTPUT TYPE FROM BMS		THERMS	ALARMS	NOTES
	HW	VP (V)		VP (V)	HW	DI	AI			
START/STOP	*					*	*	*	*	1
STATUS (CURRENT SENSOR)	*				*		*	*	*	2
TEMP SETPOINT		*				*	*	*	*	3
DA TEMP TO AN EQUIPMENT		*	*	*	*	*	*	*	*	4
ON INTAKE DAMPER			*	*	*	*	*	*	*	5
NOTE: (EXAMPLE SCENARIOS) 1. ENABLE/DISABLE SIGNAL FROM THE BMS SYSTEM. 2. MONITOR A STATUS WITH AN ANALOG INPUT TO THE BMS SYSTEM. 3. OUTPUT AN ANALOG SIGNAL THROUGH INTEGRATION FROM THE BMS SYSTEM TO AN EQUIPMENT WITH FACTORY CONTROLS. 4. READ AN ANALOG SIGNAL THROUGH INTEGRATION FROM AN EQUIPMENT WITH A FACTORY- PROVIDED / FIELD-INSTALLED SENSOR TO THE BMS SYSTEM. 5. INTERLOCK HARDWIRES (BINARY INPUT AND BINARY OUTPUT) BETWEEN AN ACTUATOR TO AN EQUIPMENT WITHOUT ANY BMS CONTROL OR MONITORING.										

TYPICAL REFERENCE FLAG / NAME REPRESENTATIONS *

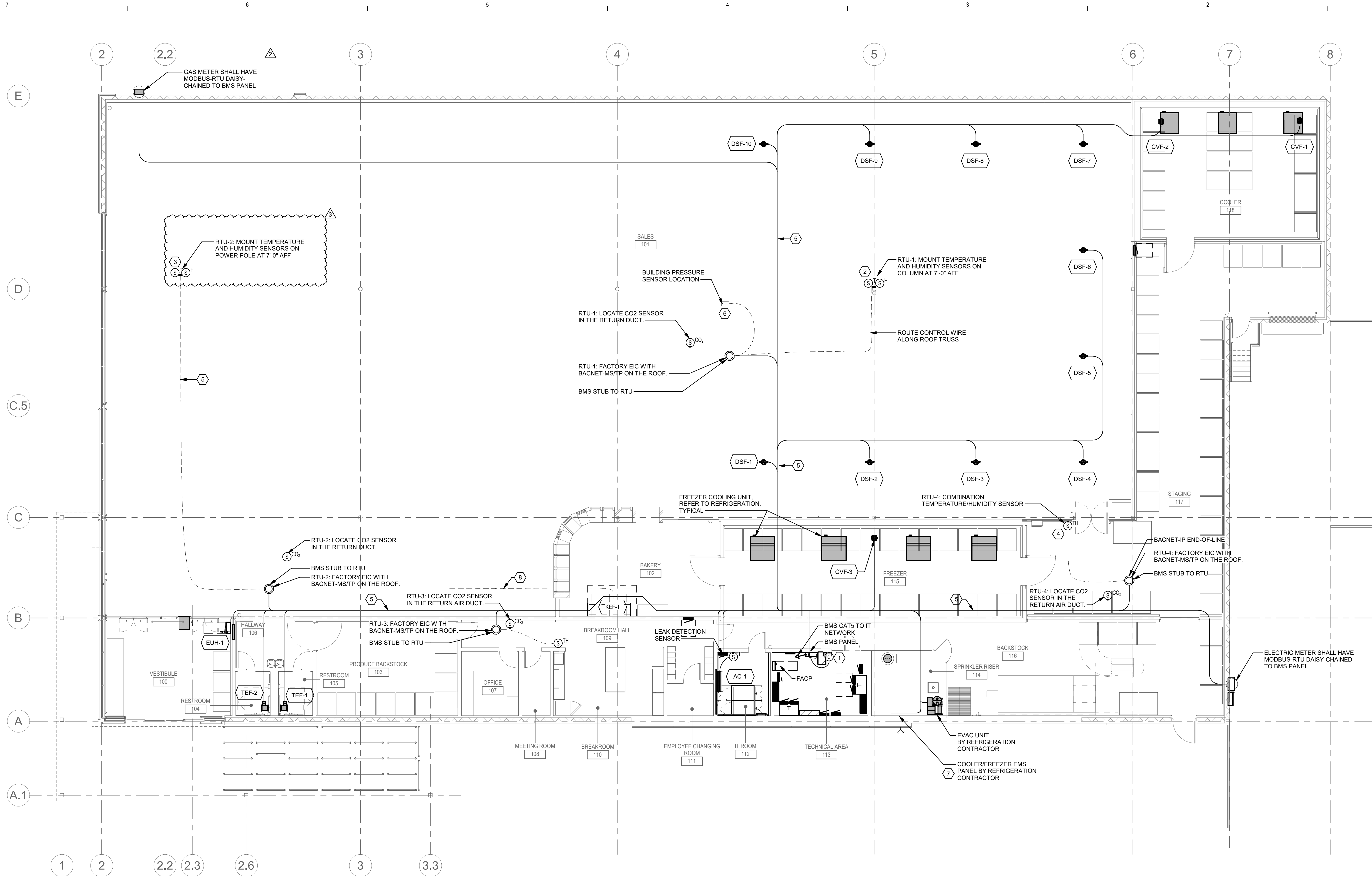
A0J	ADJUSTABLE (BY OWNER)
AAC	APPLICATION ADVANCE CONTROLLER
ASC	APPLICATION SPECIFIC CONTROLLER
ACT	ACTUATOR
AFI	ABOVE FINISH FLOOR
AFS	AIR FLOW MEASURING STATION
ALRM	ALARM
BMS	BUILDING MANAGEMENT SYSTEM
BSP	BUILDING STATIC PRESSURE
BOP	BUILDING OUTSIDE AIR
CFM	CUBIC FEET PER MINUTE
CO2	CARBON DIOXIDE
CT	CURRENT TRANSDUCER
DA-SP	DISCHARGE AIR TEMPERATURE SETPOINT
DMPR	DAMPER
DPS	DIFFERENTIAL PRESSURE SWITCH
DPT	DIFFERENTIAL PRESSURE TRANSDUCER
DX	DIRECT EXPANSION
EA	EXHAUST AIR
ECON	ECONOMIZER
EIC	EQUIPMENT INTEGRAL CONTROLLER
ENABLE	UNIT ENABLED (FAN RUNNING)
ES	END SWITCH
FA	FIELD ADJUSTABLE (BY FIELD PROGRAMMER)
F.O.	FALL OPEN VIA SPRING
F.C.	FALL CLOSE VIA SPRING
HDA	HAND-OFF-AUTO SWITCH
MIN/OA	MINIMUM OUTSIDE AIR
MOD	MODULATING SIGNAL VIA 4-20MA OR 0-10VDC
MOD-F	MODULATING SIGNAL VIA FLOATING
N.O.	NORMAL OPEN VIA SPRING
N.C.	NORMAL CLOSE VIA SPRING
OA-DM	DESIGN MINIMUM OA AIR FLOW SETPOINT
OA-CO2	OUTSIDE AIR CO2
ORH	OUTSIDE AIR RELATIVE HUMIDITY
OAT	OUTSIDE AIR TEMPERATURE
OC	OCCUPANCY MODE ENABLED
PAT	PREHEAT AROUNDCHARGE TEMPERATURE
PARG	NIGHT PURGE MODE ENABLED
PAWR	POWER
RARH	RETURN AIR RELATIVE HUMIDITY
RAT	RETURN AIR TEMPERATURE
SARH	SUPPLY AIR RELATIVE HUMIDITY
SAT	SUPPLY AIR TEMPERATURE
SD	SMOKE DETECTOR
SP-CO2	SPACE AIR CO2
SPD	SUPPLY FAN SPEED
SPRH	SPACE AIR RELATIVE HUMIDITY
SPT	SPACE AIR TEMPERATURE
VFD	VARIABLE FREQUENCY DRIVE (ALSO VSD)
WS	WATER SENSOR/DETECTION SWITCH

*NOTE: THIS IS A LIST OF TYPICAL REPRESENTATIONS. THIS LIST IS NOT INTENDED TO BE INCLUSIVE OR DEFINITIVE. REFER TO ACTUAL LOGIC DIAGRAM TO DETERMINE ACTUAL REFERENCE TAG REPRESENTATION.

CONTROL SYMBOLS

SYMBOL	DESCRIPTION
	POINT NAME
	SPACE TEMPERATURE SENSOR
	SPACE HUMIDITY SENSOR
	SPACE TEMPERATURE AND HUMIDITY SENSOR
	SPACE CO2 SENSOR
	SPACE CARBON MONOXIDE SENSOR
	LOW-VOLTAGE THERMOSTAT
	TEMPERATURE SENSOR WITH SINGLE POINT ELEMENT, DUCT-MOUNTED
	CARBON DIOXIDE SENSOR, DUCT-MOUNTED
	HUMIDITY & TEMPERATURE SENSOR, DUCT-MOUNTED
	SMOKE DETECTOR
	DIFFERENTIAL PRESSURE SWITCH
	AIR FLOW SWITCH
	DIFFERENTIAL PRESSURE TRANSMITTER/SENSOR
	WATER LEAK DETECTOR
	CONTROL TRANSFORMER
	RELAY COILS
	NORMALLY OPEN AND NORMALLY CLOSED CONTACTS
	CURRENT SWITCH / SENSOR (ON/OFF OR VFD SENSOR)
	TEMPERATURE CONTROL PANEL
	CONTROL DAMPER (OPPOSED-BLADE)
	CONTROL DAMPER (PARALLEL-BLADE)
	DAMPER ACTUATOR WITH ELECTRIC ACTUATOR
	WIRING DESIGNATION

- DRAWING NOTES:
- ALL BMSF SERIES DRAWINGS ARE BUILDING AUTOMATION SYSTEM / TEMPERATURE CONTROLS
 - REFER TO M-000 AND E-000 FOR ADDITIONAL SYMBOLS AND ABBREVIATIONS.
 - CONTROL DRAWING DETAILS ARE NOT TO SCALE.



BMS FLOOR PLAN
1/8" = 1'-0"

SPACE TEMPERATURE SETPOINTS

AREA	OCCUPIED PERIOD		UNOCCUPIED PERIOD		NOTES
	HEATING	COOLING	HEATING	COOLING	
SALES, RTU-1	68 °F (±2°F)	73 °F (±2°F)	65 °F (±2°F)	73 °F (±2°F)	1 - 2
CHECKOUT AREA, RTU-2	70 °F (±2°F)	73 °F (±2°F)	65 °F (±2°F)	78 °F (±2°F)	1 - 2
BREAKROOM & OFFICE, RTU-3	70 °F (±2°F)	79 °F (±2°F)	65 °F (±2°F)	79 °F (±2°F)	2
WAREHOUSE AND TECHNICAL ROOM, RTU-4	63 °F (±2°F)	73 °F (±2°F)	60 °F (±2°F)	78 °F (±2°F)	1 - 2
TECHNICAL AREA (IT/SERVER), AC-1	64 °F (±2°F)	75 °F (±2°F)	64 °F (±2°F)	75 °F (±2°F)	2

1. DEHUMIDIFICATION SETPOINT SHALL BE 50%RH (ADJ).
2. OCCUPIED/UNOCCUPIED PERIODS TO BE COORDINATED WITH LIDL'S REPRESENTATIVE.

GENERAL NOTES

- REFER TO BMS-400 SHEETS FOR CABLING DETAILS FOR THE FULL LIST OF FIELD INSTALLED INPUT/OUTPUT POINTS.
- THE FLOOR PLAN DETAILS PRIMARILY PROVIDE THE FOLLOWING:
 - A. BACNET-MSTP ROUTING
 - B. BACNET-IP ROUTING
 - C. RTU SPACE TEMP SENSOR LOCATIONS
 - D. RTU DUCT CO₂ SENSOR LOCATIONS
 - E. RTU SPACE HUMIDITY SENSOR LOCATIONS
 - F. AC UNIT THERMOSTAT LOCATION
 - G. CUF FAN AIRFLOW SWITCH LOCATIONS
- ALL LOW VOLTAGE CABLES SHALL BE 18 GAUGE, STRANDED, AND SHIELDED CABLES. CONDUCTOR WILL VARY.

BMS KEYNOTES

- PROVIDE MANUAL VARIABLE SPEED CONTROLLER FOR DESTRATIFICATION FANS. MANUFACTURER ARIUS MODEL TRIAC-120-5. 120V SAMPs. INSTALLED BY ELECTRICAL CONTRACTOR.
- REFER TO SHEET BMS-402 FOR SENSOR MOUNTING DETAILS.
- SENSORS TO BE MOUNTED ON THE DATA POST FACING STORE SALES AREA. REFER TO SHEET BMS-403 FOR SENSOR MOUNTING DETAILS.
- MOUNT RTU-4 COMBINATION TEMPERATURE/HUMIDITY SENSOR RECESSED IN WALL AT 9'-0" AFF.
- ROUTE BMS CONTROL LOW VOLTAGE CABLES IN CABLE TRAY. BMS CABLE SHOWN IS DIAGRAMMATIC INDICATING COMMON DEVICES INTERLOCKED TO BMS. INDIVIDUAL DAISY-CHAINED CIRCUITS TO BMS PANEL AND BAC-NET-MSTP END-OF-LINE(S) NOT SHOWN. BMS CONTRACTOR TO COORDINATE QUANTITY OF REQUIRED DAISY-CHAIN CIRCUITS WITH THE OTHER TRADES.
- INTERLOCK BUILDING AND ATMOSPHERIC PRESSURE SENSORS TO THE BMS FOR MONITORING AND DATA LOGGING PURPOSES.
- REFER TO REFRIGERATION EMS DRAWINGS FOR ADDITIONAL CABLE TERMINATION REQUIREMENTS.
- INTERLOCK EXHAUST FAN KEF-1 TO RTU-2 TO INDEX RTU-2 TO OCCUPIED MODE WHEN EVER KEF-1 IS ON.

ISSUANCE

10/16/2025	100% QA/QC
11/07/2025	PERMIT CYCLE 1
01/14/2026	PERMIT CYCLE 3
05/28/2026	CCD-01

PROJECT NUMBER
7955191
PROJECT MANAGER
S. CHOVAN
PROFESSIONAL

DRAWN BY
EBZ
CHECKED BY
KPM



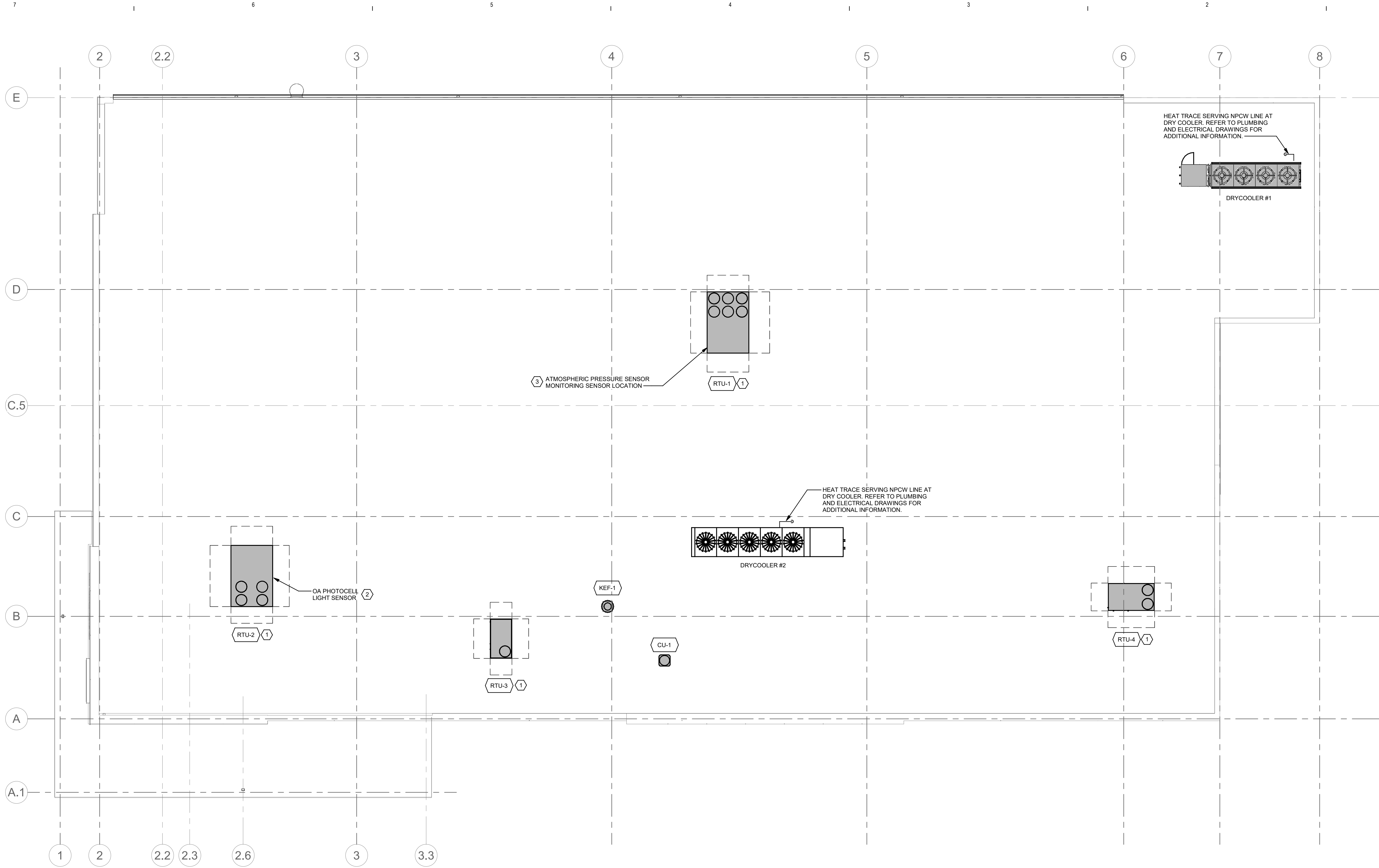
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LIDL 1517 NORCROSS, GA

499 JIMMY CARTER BLVD. NORCROSS, GA 30063



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BMS ROOF PLAN

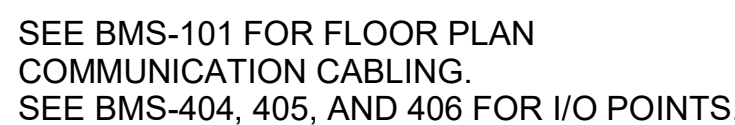
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BMS ROOF PLAN
1/8" = 1'-0"

BMS KEYNOTES (R)

1. BMS INSTALLER SHALL CABLE UTILIZING CONDUIT INSTALLED BY DIVISION 26 THROUGH THE ROOFTOP CURB. REFER TO ELECTRICAL DRAWINGS FOR ADDITIONAL INFORMATION.
2. BMS INSTALLER SHALL PROVIDE A SECURED POST ON THE ROOFTOP UNITS HOOD. REFER TO PHOTOCELL LIGHT SENSOR DETAIL ON BMS-406 AND ELECTRICAL DRAWINGS FOR ADDITIONAL INFORMATION.
3. INTERLOCK BUILDING AND ATMOSPHERIC PRESSURE SENSORS TO THE BMS FOR MONITORING AND LOGGING PURPOSES.

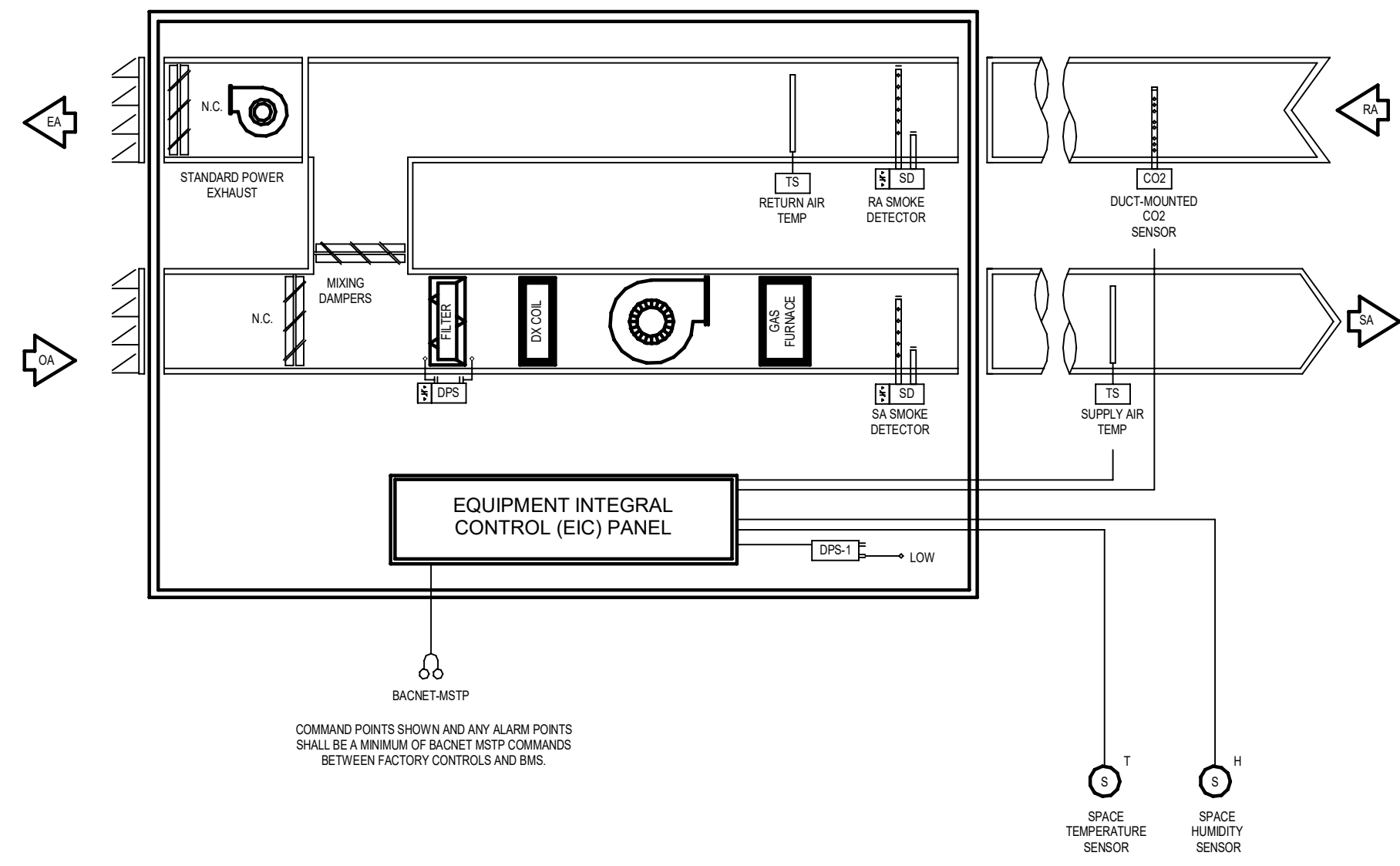


1. THE MANAGERMENT SYSTEM AND THE EDC CONTROLLERS, THE PRO CONTROLLER SHALL BE SPECIFICALLY CONFIGURED TO COMMUNICATE OVER LCL'S PROPRIETARY NETWORK PROTOCOL.
2. COPELAND SHALL PROVIDE LOCAL SOFTWARE AND ENTERPRISEWIDE SOFTWARE LICENSES TO LCL NATIONAL FOR COMPLETE OWNERSHIP OF THEIR OWN SYSTEM.
3. THE COPELAND BMS SYSTEM SHALL CONTROL AND/MONITOR THE VARIOUS ECOSYSTEMS AS SHOWN IN THE BMS SHEETS.
4. THE BMS INSTALLER SHALL BE RESPONSIBLE FOR INSTALLING THE BMS PANEL AND TERMINATING THE 10 POINT SPECIFIC TO THE I/O SCHEDULE AS SHOWN ON BMS-40A, 42S, AND 40K SHEETS.
5. GRAPHICAL USER INTERFACE (GUI) COPELAND SHALL PROVIDE A CUSTOM GUI WITH ADJUSTABLE TWIN-KEYS READ/WRITE CAPABILITIES.
6. COPELAND SHALL PROVIDE THE SAME KEY CONFIGURATIONS FOR ALL BMS PANELS AND TWIN-KEYS (TWO KEYS SHALL BE INCLUDED). IF A KEY IS MISSING, COPELAND SHALL PROVIDE NEW LOGS AND KEYS TO ALL ENCLOSURES AT NO COST TO THE OWNER.

BMS NETWORK LAYOUT

BMS-401

01 ROOFTOP UNIT (RTU-1, SALES): CONTROL SCHEMATIC, CONTROLS BILL-OF-MATERIAL, & SEQUENCE

BILL-OF-MATERIAL
RTU-1 CONTROL DEVICES ONLY

NO.	DEVICE	QTY	PROVIDED BY		CONDUIT		CABLING		DEVICE	MODEL NO.	NOTES
			LENNOX	COPELAND	DIV. 26	DIV. 26	BMS INSTALLER	DIV. 26			
1	OUTDOOR AIR TEMP SENSOR	1	•								1
2	SUPPLY AIR TEMP SENSOR	1	•								
3	RETURN AIR TEMP SENSOR	1	•								1
4	SUPPLY AIR SMOKE DETECTOR	1	•								1
5	RETURN AIR SMOKE DETECTOR	1	•								1
6	DUCT CO2 SENSOR	1	•						LENNOX	CD0005S4E11	
7	SPACE TEMP SENSOR	1	•						LENNOX	BATRK2LWPV-10	8
8	SPACE HUMIDITY SENSOR	1	•						LENNOX	ITM20	7
9	ROOM STATIC PRESSURE PICKUP	1	•						KELE	RPS	2
10	1/4" POLY TUBE (100 FEET)	1	•						KELE	T-101	
11	DIFFERENTIAL PRESSURE SENSOR	1	•						BETRA	2011MRWD	3.4
12	SECONDARY CONTROLLER	1	•						COPELAND	MATPLEX 1840	3

NOTE:
1. FACTORY MOUNTED AND WIRED INTERNALLY TO THE UNIT.
2. COLOR SHALL BE WHITE.
3. INSTALL IN ROOFTOP UNIT CONTROL CABINET.
4. PRESSURE SENSOR SHALL BE SET FOR 0.05 INCHES OF WATER.
5. PROVIDE CONTROLLER AND 24VAC TRANSFORMER.
6. WHITE BUZZER STYLE TEMPERATURE SENSOR SHALL BE PROVIDED.
7. POLY TUBE MOUNTED ON EXTENSIBLE MOUNTED POINT JUST ABOVE SHELVING ROUGHLY 7' A.F.F.

THE DETAILS AND SEQUENCE IS BASED ON LENNOX UNIT BASED OF DESIGN. IF AN ALTERNATE UNIT IS PURCHASED, IT IS THE RESPONSIBILITY OF THE EQUIPMENT PURCHASING CONTRACTOR AND CONTROL CONTRACTOR TO PROVIDE CONTROL SEQUENCES, ALL POINTS, DEVICES, ETC FOR A FUNCTIONAL UNIT AT NO COST TO THE DESIGNER AND OWNER.

OPERATING PARAMETER OVERVIEW

ALL PARAMETER VALUES SHALL BE MADE ADJUSTABLE (ADJ) BY THE OWNER AT THE BMS GRAPHICAL INTERFACE.

ALTERNATE OPERATIONAL SCHEDULES:
UNOCCUPIED: WHEN BURGLAR ALARM IS TRIGGERED
OCCUPIED: WHEN BURGLAR ALARM IS TRIGGERED

PARAMETERS:
OCCUPIED SPACE TEMP SETPOINT: SEE SHEET BMS-101, CONTROL SETPOINT TABLE.
UNOCCUPIED SPACE TEMP SETPOINT: SEE SHEET BMS-101, CONTROL SETPOINT TABLE.
DEHUMIDIFICATION SETPOINT: SEE SHEET BMS-101, CONTROL SETPOINT TABLE.

CO2 SETPOINT: 1,000 PPM (ADJ)
DFF PRESSURE SENSOR NO. 1 (DPS-1): BMS MONITOR AND LOG

RTU-1 POINTS LIST
(BACNET OBJECTS)

POINT NAME	BMS SIGNAL TYPE	FACTORY POINT	INPUT TO BMS	OUTPUT FROM BMS	GRAPHICS	TRENDS	ALARM	NOTES
LENNOX BACNET POINT								
SUPPLY FAN RUNNING								
SUPPLY FAN RUN TIME								
SUPPLY FAN ERROR								5
COMPRESSOR RUNNING								
COMPRESSOR RUN TIME								
COMPRESSOR ALARM								6
HEATING RUNNING								
HEATING RUN TIME								
HEATING ERROR								7
ECONOMIZER RUNNING								
ECONOMIZER RUN TIME								
ECONOMIZER ALARM								
SUPPLY AIR TEMP								1
SPACE TEMP								1.2
SPACE TEMP ACTUAL SETPOINT								
SPACE TEMP OCC CLG SETPOINT								
SPACE TEMP OCC HTG SETPOINT								
SPACE TEMP UNOCC CLG SETPOINT								
SPACE TEMP UNOCC HTG SETPOINT								
DUCT RETURN AIR CO2								1.3,8
DUCT RETURN AIR CO2 SETPOINT								8
DPS-1 SENSOR								1
PWR EXH. DPS SETPOINT								6
DND DPS SETPOINT								8
PWR EXHAUST FAN RUN PROOF								6
SPACE REL. HUMIDITY								1
SPACE RH DEHUM. SETPOINT								4
OUTDOOR AIR TEMP								
OCC SCHEDULE ENABLE								
CURRENT ERROR CODE								

NOTE:
1. FACTORY PROVIDED. BMS INSTALLER SHALL FIELD INSTALL.
2. BMS SHALL PLACE SOFTWARE HIGH LIMIT (150 PPM (ADJ)) FOR ALARMING.
3. BMS SHALL PLACE SOFTWARE HIGH LIMIT (150 PPM (ADJ)) FOR ALARMING.
4. BMS SHALL PLACE SOFTWARE HIGH LIMIT (150 PPM (ADJ)) FOR ALARMING.
5. LENNOX BACNET ALARM ID 1, 14, 15, AND 16 SHALL ALARM "SUPPLY FAN ERROR".
6. LENNOX BACNET ALARM ID 1, 14, 15, AND 16 SHALL ALARM "COMPRESSOR ERROR".
7. LENNOX BACNET ALARM ID 1, 14, 15, AND 16 SHALL ALARM "HEATING FAN".
8. POINTS LISTED HEREIN ARE FOR THE CONTROL AND SHALL BE USED FOR BMS BOARD.

GENERAL POINTS LIST

WIRING BY DIVISION 26 CONTRACTOR
WIRING BY BMS INSTALLER
WIRING BY DIVISION 26 CONTRACTOR (WIRING INCLUDING TERMINATION) BY BMS INSTALLER
FIELD WIRING (TYPICAL)
BMS CALL OUT NOTE (TYPICAL)

GENERAL NOTES

BMS INTERFACE LOGIC

- SEQUENCE OF OPERATION - ECONOMIZER AND CO2 SENSOR
- THE ROOFTOP UNIT FACTORY PROVIDED SECONDARY CONTROLLER SHALL BE POWERED THROUGH A FACTORY PROVIDED AND WIRED 24VAC TRANSFORMER. THE TRANSFORMER SHALL PROVIDE 24VAC POWER TO THE SPACE TEMPERATURE SENSOR, HUMIDITY SENSOR, DUCT MOUNTED CO2 SENSOR, AND SPACE MOUNTED DPS-1 SENSORS.
 - THE SECONDARY CONTROLLER SHALL BE NETWORKED TO THE BMS SYSTEM THROUGH BACNET-MSTR BY THE BMS INSTALLING CONTRACTOR.
 - A DUCT MOUNTED CO2 SENSOR, HUMIDITY SENSOR, AND DIFFERENTIAL PRESSURE SENSOR (DPS-1) SHALL BE FIELD CONNECTED TO THE SECONDARY CONTROLLER BY THE BMS INSTALLING CONTRACTOR.
 - AN ANALOG OUTPUT ON THE SECONDARY CONTROLLER SHALL BE CABLED TO THE POWER EXHAUSTS DPS INPUT POINT BY THE BMS INSTALLING CONTRACTOR.
 - AN ANALOG OUTPUT ON THE SECONDARY CONTROLLER SHALL BE CABLED TO THE POWER EXHAUSTS DPS INPUT POINT BY THE BMS INSTALLING CONTRACTOR.
 - SEQUENCE OVERVIEW NO. 1: A CO2 SENSOR SHALL ALLOW CONTROL OF RTU-1'S OUTDOOR DAMPER, ACCOMPLISHED THROUGH AN ANALOG OUTPUT FROM THE SECONDARY CONTROLLER TO THE PRIMARY CONTROLLER. THE PRIMARY CONTROLLER SHALL SUBSEQUENTLY MODULATE THE OUTDOOR AIR DAMPER OPEN TO ITS MINIMUM SETPOINT POSITION WHEN SPACE CO2 LEVEL EXCEEDS SETPOINT. SEE ROOFTOP UNIT-200-001 FOR FURTHER INFORMATION ON DAMPER OPEN QUANTITY.
 - SEQUENCE OVERVIEW NO. 2: THE OUTSIDE AIR TEMPERATURE SENSOR SHALL ALLOW CONTROL OF RTU-1'S OUTDOOR AIR DAMPER, ACCOMPLISHED THROUGH AN ANALOG OUTPUT FROM THE SECONDARY CONTROLLER TO THE PRIMARY CONTROLLER. THE PRIMARY CONTROLLER SHALL SUBSEQUENTLY MODULATE THE OUTDOOR AIR DAMPER FULLY OPEN WHEN AMBIENT AIR TEMPERATURE IS FAVORABLE FOR PROVIDING FREE COOLING VIA AN ECONOMIZER MODE. THE POWER EXHAUST FAN TO RUN WHEN OUTDOOR AIR INTAKE DAMPERS POSITION IS 80% (ADJ) OPEN OR MORE.

SEQUENCE OF OPERATION - BUILDING PRESSURE MONITOR SENSOR

- SEQUENCE OF OPERATION - BUILDING PRESSURE MONITOR SENSOR
- THE BUILDING AND ATMOSPHERIC PRESSURE SENSORS TO THE BMS FOR MONITORING AND DATA LOGGING PURPOSES SHALL BE NETWORKED TO THE BMS SYSTEM THROUGH BACNET-MSTR BY THE BMS INSTALLING CONTRACTOR.

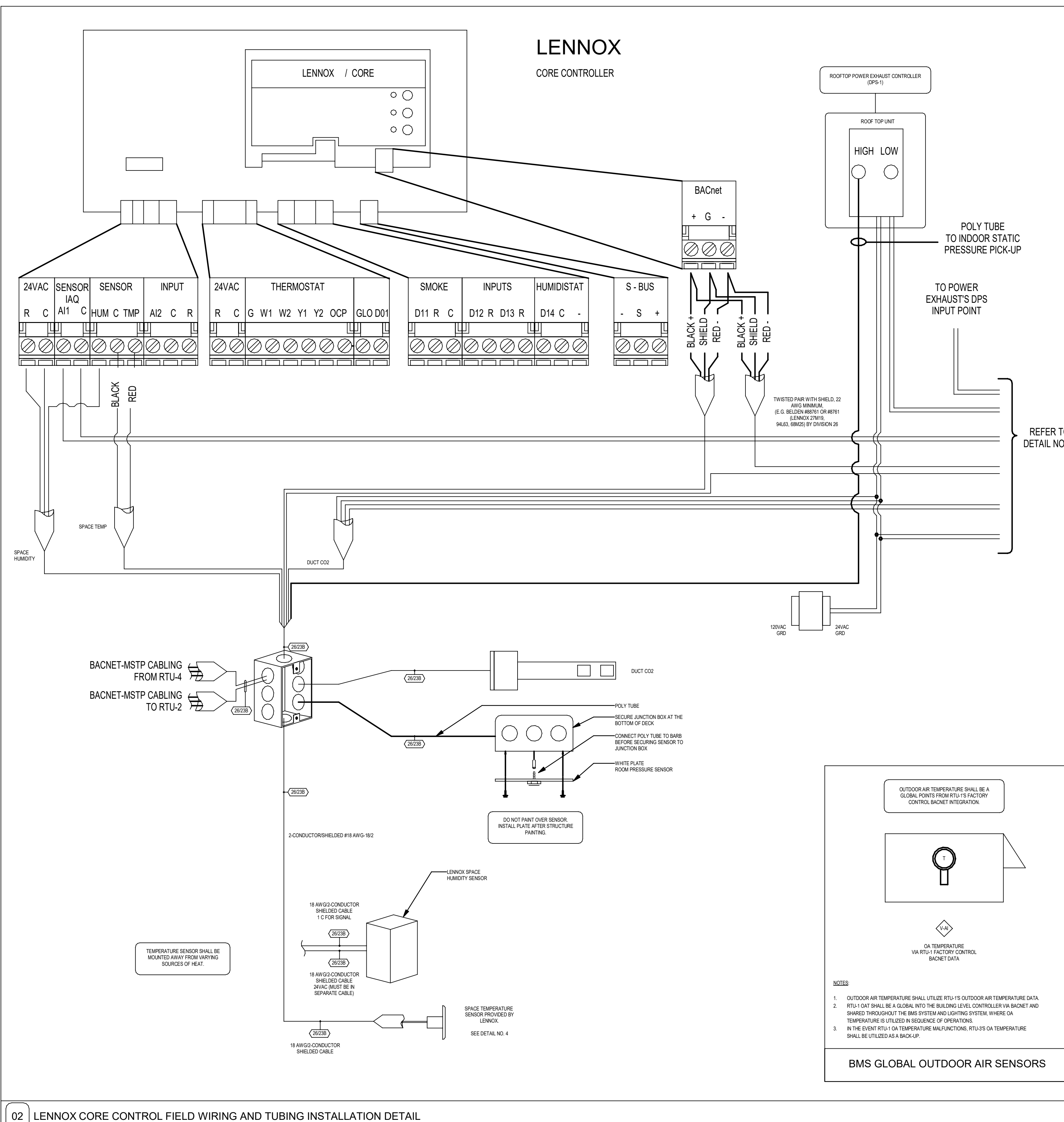
SEQUENCE OF OPERATION

- SYSTEM OPERATING MODES:
- THE SYSTEM CONTROLLER SHALL SEND THE EQUIPMENT CONTROLLERS OCCUPIED/UNOCCUPIED, MORNING WARMUP/COOL, AND NIGHT/COOL MODES. IF COMMUNICATION LOST, THE EQUIPMENT CONTROLLERS SHALL OPERATE USING MODES AND SETPOINTS.
 - NIGHT SETBACK DURING UNOCCUPIED MODE: THE SYSTEM SHALL SHUT OFF IF THE ZONE TEMPERATURE DRIFTS TO THE UNOCCUPIED HEATING/COOLING SETPOINT. THE SYSTEM SHALL START UP TO HEAT OR COOL THE ZONE, WHILE THE OUTDOOR AIR REMAINS CLOSED UNLESS ECONOMIZING.
 - OPTIMAL START: THE SYSTEM CONTROLLER SHALL AUTOMATICALLY DETERMINE THE OPTIMAL START TIME, SUCH THAT EACH ZONE REACHES ITS OCCUPIED SETPOINT AS EARLY AS POSSIBLE FOR SCHEDULED OCCUPANCY.
 - DEMAND CONTROLLED VENTILATION: EQUIPPED WITH CO2 SENSOR, OUTDOOR AIRFLOW SHALL BE RESET BASED ON OCCUPANCY STATUS AND/OR MEASURED CO2 CONCENTRATION.
 - OCCUPIED HEAT/COOL: THE RTU SHALL OPERATE THE SUPPLY FAN CONTINUOUSLY AND MODULATE (OR CYCLE) COMPRESSORS, MODULATE (OR STAGE) HEAT, AND/OR ECONOMIZER ARRANGING ECONOMIZING TO MAINTAIN ZONE TEMPERATURE AT SETPOINT. THE OUTDOOR AIR SHALL OPEN TO BRING IN THE REQUIRED AMOUNT OF VENTILATION.
 - MORNING WARMUP/COOL: THE RTU SHALL OPERATE THE SUPPLY FAN AND MODULATE (OR CYCLE) COMPRESSORS OR MODULATE (OR STAGE) HEAT TO RAISE LOWER ZONE TEMPERATURE TO ITS OCCUPIED SETPOINT. THE OUTDOOR AIR SHALL REMAIN CLOSED UNLESS ECONOMIZING.
 - SMOKE DETECTOR: ROOFTOP UNIT SHALL SHUTDOWN UPON SMOKE OR RA SMOKE DETECTOR SENSING SMOKE AND SHALL SEND A SIGNAL TO THE FIRE ALARM PANEL.

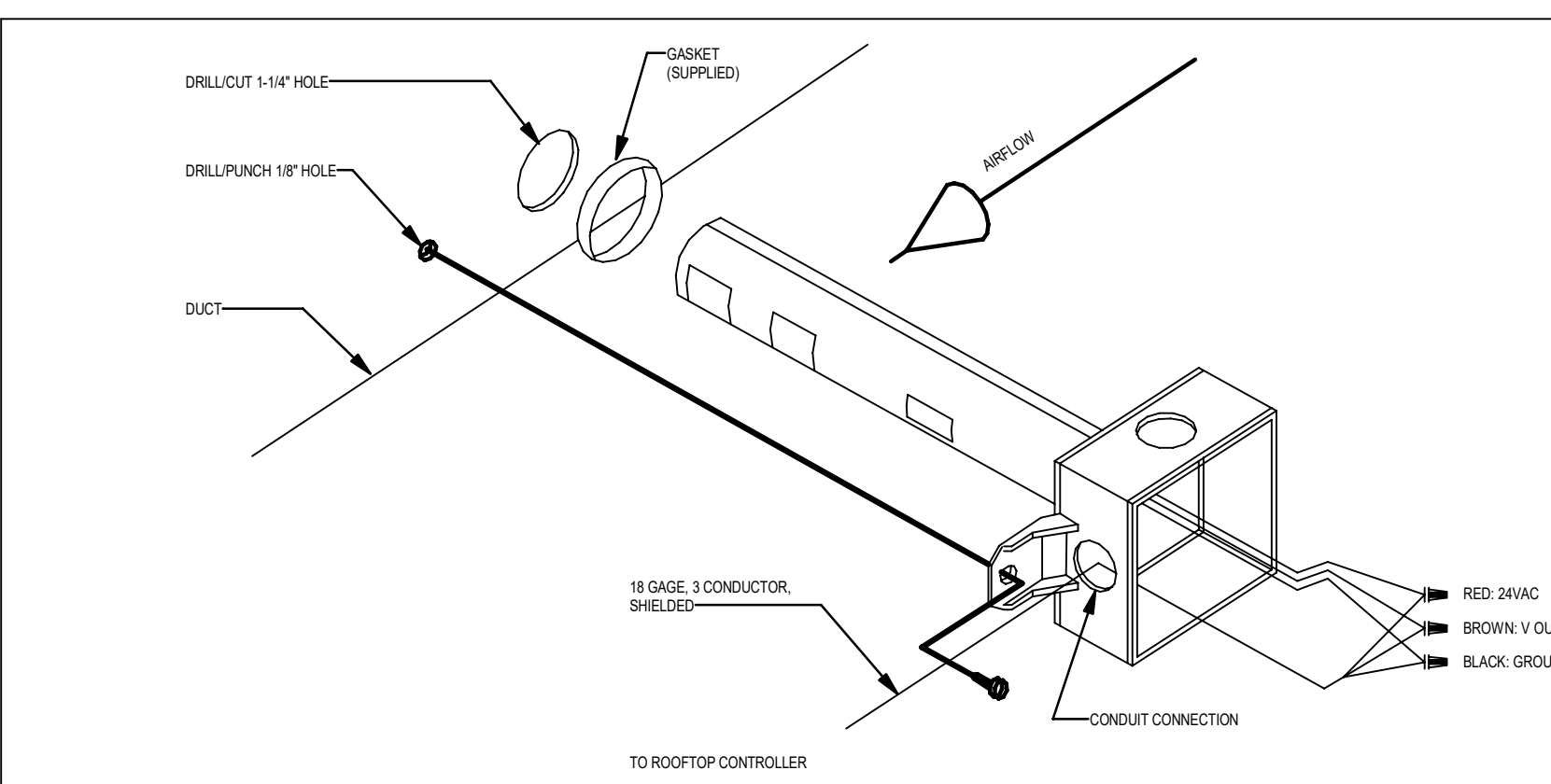
SINGLE ZONE VAV SYSTEM

- SINGLE ZONE VAV SYSTEM
- OCCUPIED HEAT/COOL: THE RTU SHALL MODULATE THE SUPPLY FAN, MODULATE (OR CYCLE) COMPRESSORS, MODULATE (OR STAGE) HEAT, AND/OR ECONOMIZER ARRANGING ECONOMIZING TO MAINTAIN ZONE TEMPERATURE AT SETPOINT. THE OUTDOOR AIR SHALL MODULATE IN PROPORTION TO CHANGING SUPPLY FAN SPEED, TO BRING IN THE REQUIRED AMOUNT OF VENTILATION.
 - MORNING WARMUP/COOL: THE RTU SHALL OPERATE THE SUPPLY FAN AND MODULATE (OR CYCLE) COMPRESSORS OR MODULATE (OR STAGE) HEAT TO RAISE LOWER ZONE TEMPERATURE TO ITS OCCUPIED SETPOINT. THE OUTDOOR AIR SHALL REMAIN CLOSED UNLESS ECONOMIZING.

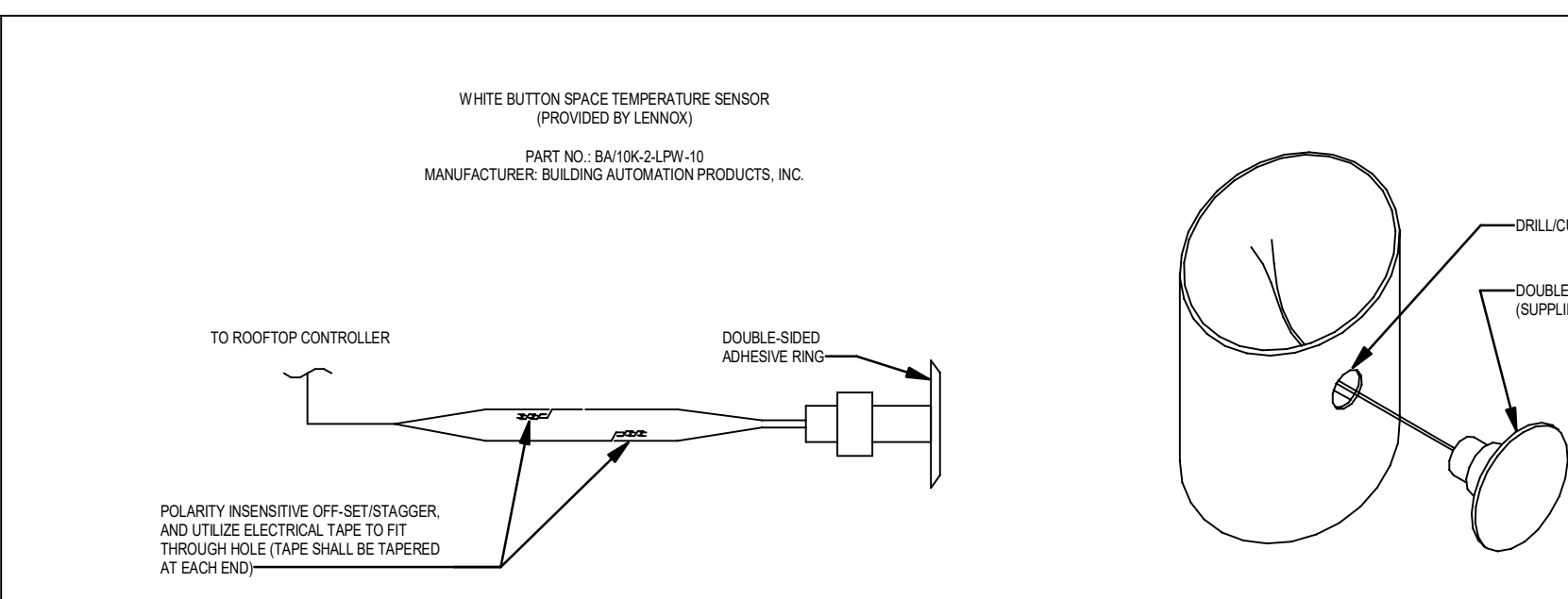
02 LENNOX CORE CONTROL FIELD WIRING AND TUBING INSTALLATION DETAIL



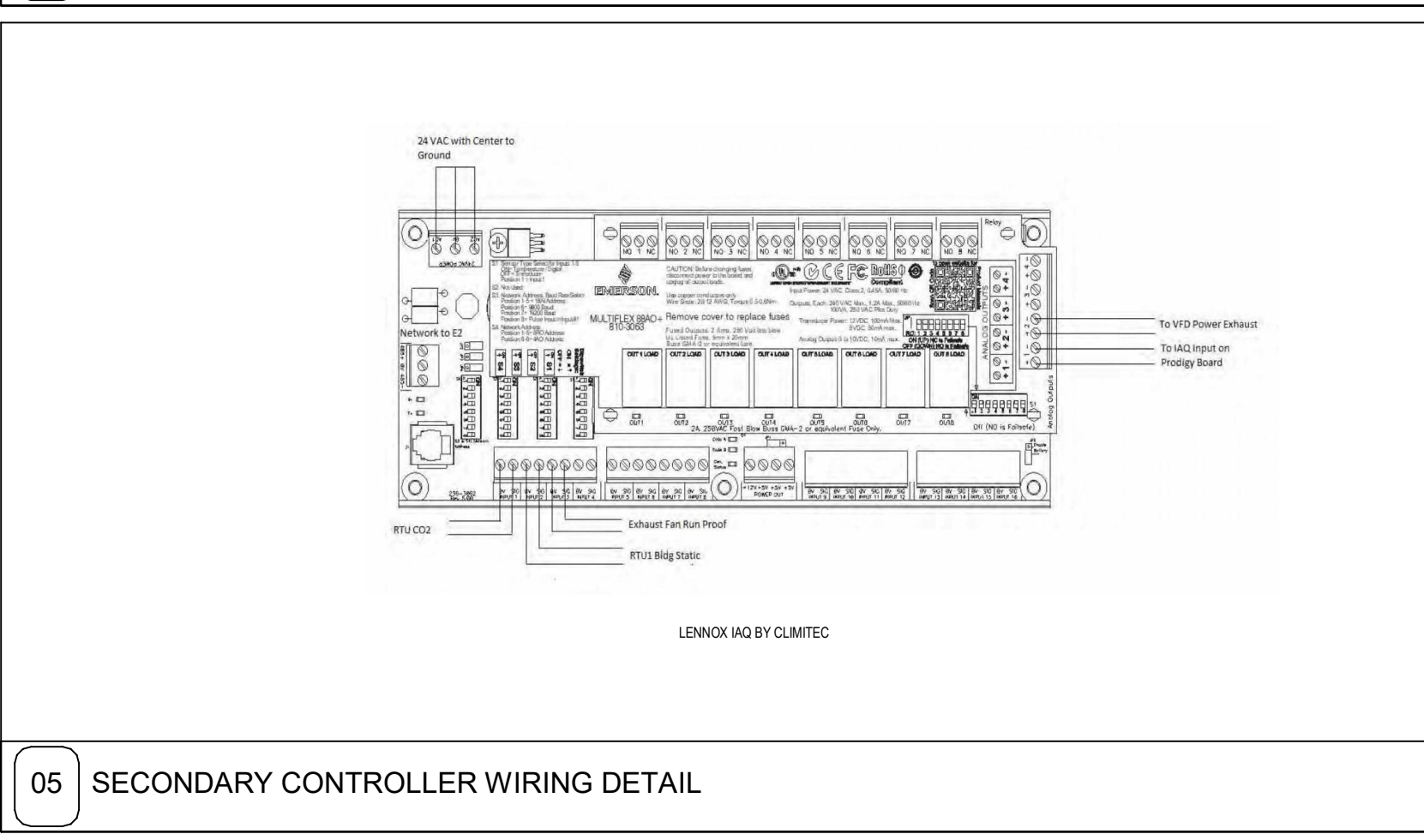
03 DUCT CO2 CONTROL WIRING DETAIL

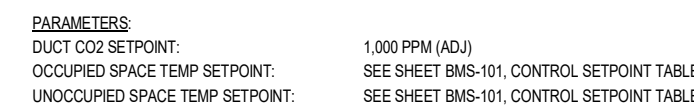


04 SPACE COMBINATION TEMPERATURE & HUMIDITY SENSOR WIRING DETAIL



05 SECONDARY CONTROLLER WIRING DETAIL





06 SPACE HUMIDITY SENSOR WIRING DETAIL - BTU-2

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